DiseaseSpread: Multi-Factor Outbreak Risk Scoring System

Project Report

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United Nations Sustainable Development Goal Alignment

SDG Goal 3: Good Health and Well-Being

Abstract

Disease outbreaks continue to pose a significant threat to public health, particularly in densely populated and resource-constrained regions like India. Traditional disease surveillance systems are largely reactive, triggering responses only after outbreaks have already taken hold. This leaves health authorities with insufficient lead time to allocate resources or mount effective preventive interventions.

This project, DiseaseSpread, addresses this limitation by developing a Multi-Factor Outbreak Risk Scoring (ORS) system that uses historical epidemiological data from India's Integrated Disease Surveillance Programme (IDSP), combined with climate variables such as temperature, rainfall, and leaf area index, along with population density metrics drawn from the Census of India 2011. The project focuses specifically on Tamil Nadu, analyzing district-level data across 25 districts.

The system follows a structured data science pipeline covering data collection, preprocessing and feature engineering, exploratory data analysis, visualization, and predictive modeling. An Outbreak Risk Score is computed for each district by assigning weighted contributions to case history, rainfall intensity, temperature, and population density. A Random Forest classifier, trained with SMOTE-based oversampling to handle class imbalance, is then applied to forecast disease case surges over a one-month prediction horizon.

Results are presented through choropleth-style rankings, time-series plots, and feature importance charts. The system is designed to support public health authorities in proactive, data-driven decision-making, directly contributing to the United Nations Sustainable Development Goal 3: Good Health and Well-Being.

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1. Introduction

Infectious disease outbreaks are among the most pressing challenges faced by public health systems worldwide. In a country as densely populated as India, where climate variability is high and healthcare infrastructure is unevenly distributed, the ability to predict and preempt disease surges before they escalate can save thousands of lives and prevent enormous economic loss.

The Integrated Disease Surveillance Programme (IDSP), operated under India's Ministry of Health and Family Welfare, provides weekly district-level reports on disease incidence across a range of communicable diseases including dengue, malaria, typhoid, cholera, and acute respiratory infections. While this data provides valuable historical insight, its primary use has been retrospective rather than predictive. Converting it into an actionable forward-looking tool requires integration with environmental and demographic factors that are known to influence disease transmission.

Climate plays a well-established role in the spread of vector-borne and waterborne diseases. Rainfall patterns directly influence the breeding grounds of mosquito populations that transmit malaria and dengue fever. Temperature affects pathogen viability, vector survival rates, and the pace at which human immune responses engage. The leaf area index, a measure of plant foliage density derived from satellite data, serves as an indirect proxy for ecological conditions conducive to vector proliferation. When combined with district-level population density, these variables form a coherent multi-factor risk profile for each geographic unit.

This project, DiseaseSpread, seeks to synthesize these variables into a unified Outbreak Risk Scoring system. Using data from Tamil Nadu as a case study, the system ranks districts by their computed risk level and uses a machine learning model trained on historical surge events to forecast whether a surge is likely in the following month. The overarching goal is to transition disease surveillance from a passive, reporting-oriented function to a proactive, prediction-oriented tool that guides resource allocation and early response.

1.1 Scope

The scope of this project is limited to Tamil Nadu, India, covering 25 districts. Disease data is drawn from the IDSP surveillance bulletin records. Climate data is sourced from the ERA5 reanalysis dataset published by the Copernicus Climate Data Store. Demographic data is drawn from the Census of India 2011. The modeling horizon is a one-month lookahead forecast for binary surge classification.

1.2 Organization of the Report

This report is organized as follows. Chapter 2 states the problem in formal terms. Chapter 3 outlines the objectives. Chapter 4 provides a review of relevant literature. Chapters 5 through 9 describe the technical implementation, covering datasets, preprocessing, analysis, and modeling. Chapters 10 and 11 present results and visualizations. Chapters 12 and 13 address limitations, future directions, and conclusions. References are listed at the end.

2. Problem Statement

Traditional disease surveillance systems in India operate on a reactive model. Data collected through weekly IDSP bulletins is analyzed after incidents are already being reported, which means that by the time actionable intelligence is available, significant community spread may have already occurred. This lag between data collection and decision-making has been identified as a systemic vulnerability in outbreak response planning.

Furthermore, existing surveillance tools rarely account for the multi-causal nature of disease outbreaks. Diseases like dengue and malaria do not emerge in isolation; their incidence is correlated with monsoon intensity, standing water levels, population crowding, and temperature. A system that relies solely on reported case counts misses the predictive signal embedded in environmental and demographic data.

The core problem addressed by this project can be stated as follows: Given historical disease incidence data, climate variables, and demographic information at the district level in Tamil Nadu, is it possible to compute a reliable Outbreak Risk Score for each district and train a machine learning model to predict whether a case surge will occur in the following month?

A successful solution must satisfy three requirements. First, it must integrate heterogeneous data sources into a coherent, analysis-ready dataset. Second, it must produce an interpretable risk score that ranks districts based on their composite vulnerability. Third, it must train a classifier that achieves meaningful predictive performance on a binary surge detection task, with a particular emphasis on recall — the system's ability to detect true surge events — given the high cost of false negatives in a public health context.

3. Objectives

The objectives of this project are organized at three levels: data, analytical, and modeling.

3.1 Data Objectives

- Collect and integrate epidemiological data from IDSP weekly bulletins covering Tamil Nadu districts.
- Incorporate ERA5 climate variables including temperature, precipitation, and leaf area index.
- Merge district-level population and area data from the Census of India 2011 to compute population density.
- Produce a unified, cleaned, and analysis-ready dataset saved as merged_clean.csv.

3.2 Analytical Objectives

- Perform exploratory data analysis to understand the distribution of disease cases across districts, seasons, and years.
- Quantify the Pearson correlation between climate variables and disease case counts to identify statistically significant relationships.
- Compute a district-level Outbreak Risk Score (ORS) using a weighted multi-factor formula.
- Detect surge events in historical data using statistical thresholds.

3.3 Modeling Objectives

- Construct a binary surge classification label based on one-month lookahead case counts.
- Train a Random Forest classifier with SMOTE oversampling to address class imbalance.
- Evaluate model performance using accuracy, ROC-AUC, and classification report metrics.
- Generate and visualize feature importance scores to identify the most influential predictors.
- Save the trained model as a deployable artifact for future inference.

4. Literature Review

The intersection of epidemiology, climate science, and machine learning has produced a growing body of research on data-driven disease surveillance. This chapter reviews the foundational work and recent advances that inform the approach taken in this project.

4.1 Climate-Disease Relationships

The influence of climate on infectious disease has been documented extensively. Patz et al. (2005) demonstrated that vector-borne diseases such as malaria and dengue are highly sensitive to temperature and rainfall patterns, with transmission windows expanding under warmer, wetter conditions. In the Indian context, studies by Bhatt et al. (2013) on the global distribution of dengue fever confirmed that monsoon-associated humidity and temperature regimes in South Asia create conditions particularly favorable for Aedes aegypti mosquito proliferation, the primary vector for dengue.

Precipitation is frequently cited as the strongest single environmental predictor of acute outbreak risk. A systematic review by Stanke et al. (2013) identified floods as the leading climate-related driver of waterborne disease outbreaks including cholera, typhoid, and hepatitis A, all of which appear in the IDSP surveillance data used in this project. The inclusion of rainfall_mm as a high-weight feature (0.25) in the ORS formula is directly motivated by this evidence base.

4.2 Machine Learning in Epidemiological Forecasting

Machine learning methods have been applied to disease forecasting with increasing frequency since the early 2010s. Chretien et al. (2014) applied logistic regression and ensemble methods to influenza data and demonstrated that data-driven models could outperform traditional threshold-based surveillance alerts. Brownstein et al. (2008) pioneered the use of internet-based data for outbreak detection, showing that multi-source data integration consistently outperformed single-source models.

Random forests, the core algorithm used in DiseaseSpread, have been successfully applied to malaria risk mapping by Bhatt et al. (2015), who demonstrated that the model's ability to capture non-linear interactions between climate variables and

incidence rates made it particularly well-suited to heterogeneous epidemiological data. The model's built-in feature importance scores also make it interpretable in a way that matters for policy-level communication.

4.3 Outbreak Risk Scoring Approaches

Composite risk scoring frameworks have been used in epidemiology to aggregate multiple risk factors into a single actionable index. The WHO's Global Outbreak Alert and Response Network (GOARN) uses a multi-criteria assessment that weights epidemiological, biological, and contextual factors. More data-driven approaches include the work of Lowe et al. (2014) on dengue early warning in Brazil, which combined climate indices with disease surveillance data to generate seasonal risk scores at the municipal level.

The ORS formula in this project follows this composite weighting tradition, assigning contributions of 40 percent to case history, 25 percent to rainfall, 20 percent to temperature, and 15 percent to population density. These weights are informed by the literature cited above and calibrated against the Tamil Nadu context, where rainfall-associated vector breeding is the dominant transmission pathway for the diseases tracked in IDSP data.

4.4 Handling Class Imbalance in Health Data

Class imbalance is a pervasive challenge in epidemiological modeling, where surge events are by definition rare relative to normal surveillance periods. Chawla et al. (2002) introduced the Synthetic Minority Oversampling Technique (SMOTE), which generates synthetic samples in the feature space of the minority class rather than simply duplicating existing samples. Subsequent evaluations have confirmed SMOTE as a standard approach for improving classifier recall on minority-class health outcomes. Its use in this project directly addresses the imbalance between surge and non-surge observations in the training data.

5. Dataset Description

This project integrates four primary data sources. Each source contributes a distinct category of variables to the merged dataset, and their integration is managed through the preprocessing pipeline described in Chapter 7.

5.1 IDSP Weekly Bulletins

The Integrated Disease Surveillance Programme, administered by the Ministry of Health and Family Welfare, Government of India, publishes weekly bulletins reporting district-level disease incidence across India. The EpiClim dataset used in this project is derived from these bulletins and contains records spanning multiple years and states. Each record corresponds to a specific disease, district, and reporting week, and includes the number of confirmed cases, number of deaths, and the date of the reporting week.

For this project, records are filtered to retain only those from Tamil Nadu (identified by the state_ut column containing the substring 'Tamil'). The filtered dataset contains observations for a range of communicable diseases including dengue, malaria, typhoid, cholera, and acute respiratory infections. The key variables extracted from this source are: week_of_outbreak, state_ut, district, Disease (renamed to disease), Cases (renamed to cases), Deaths (renamed to deaths), day, mon, year, Latitude, and Longitude.

5.2 ERA5 Climate Data

Climate variables are sourced from the ERA5 global reanalysis dataset published by the European Centre for Medium-Range Weather Forecasts (ECMWF) via the Copernicus Climate Data Store. ERA5 provides gridded estimates of atmospheric variables at hourly resolution, which are aggregated to weekly district-level means for alignment with the IDSP reporting cadence.

The variables included in the EpiClim dataset from this source are: temperature (reported in Kelvin as Temp, converted to Celsius as temp_celsius by subtracting 273.15), precipitation (reported as preci in the raw data, renamed to rainfall_mm), and leaf area index (reported as LAI, renamed to leaf_area_index). The leaf area index is a

satellite-derived measure of vegetation density and serves as a proxy for ecological conditions that favor mosquito breeding.

5.3 Census of India 2011

Population and administrative area data are drawn from the Census of India 2011, published by the Office of the Registrar General and Census Commissioner of India. For this project, data for 25 Tamil Nadu districts is hardcoded into the preprocessing module as a Python dictionary, capturing district name, total population, and geographic area in square kilometers.

From these two values, population density (density_per_sqkm) is computed as population divided by area. This variable captures the demographic pressure that amplifies disease transmission: in dense urban districts like Chennai and Coimbatore, a single infectious case has a higher expected reproduction number than in sparsely populated districts like The Nilgiris.

5.4 Summary Table: Data Sources

Dataset	Source	Key Variables	Coverage
EpiClim (IDSP)	mohfw.gov.in	cases, deaths, disease, district, date	India-wide, multi-year
ERA5 Climate	Copernicus CDS	temp, rainfall, leaf area index	District-level, weekly
Census 2011	censusindia.gov.in	population, area, density	25 TN districts
WHO Disease Outbreak News	who.int	Historical outbreak records	Global

6. System Architecture and Methodology

The DiseaseSpread system is organized as a sequential data pipeline with two downstream branches: one producing the analytical Outbreak Risk Score, and the other producing a trained machine learning classifier. This chapter describes the overall architecture and the rationale behind key design decisions.

6.1 Pipeline Overview

The data pipeline begins with the raw EpiClim CSV file and proceeds through the following stages in order: data loading and state filtering, column renaming and unit conversion, missing value imputation, census data merging, case count normalization, and feature engineering. The output of this pipeline is a single consolidated CSV file, merged_clean.csv, which serves as the input for both the analytical and modeling components.

The pipeline is implemented in Preprocessing.py as a single function, run_pipeline(), which chains the six processing steps and saves the output. Each step is implemented as an independent function, enabling modular testing and reuse. The function accepts the path to the raw EpiClim CSV as input and returns the processed DataFrame.

6.2 Analysis Branch

The analysis branch, implemented in Analysis.py, operates on the merged_clean.csv output and provides five analytical functions. The summary_statistics() function aggregates total and mean case counts, maximum single-week cases, total deaths, and average temperature and rainfall by district. The disease_breakdown() function provides aggregate statistics for each disease type. The seasonal_summary() function orders seasons in the standard Tamil Nadu meteorological sequence: Winter, Summer, Monsoon, and Post-Monsoon.

The climate_disease_correlation() function computes Pearson correlation coefficients and associated p-values between each climate variable (temperature, rainfall, leaf area index, and population density) and reported case counts. The detect_surges() function applies a statistical threshold — mean plus two standard deviations per district — to flag individual records as surge events. The top_risk_districts() function invokes

compute_ors() from the preprocessing module to rank districts by their composite risk score.

6.3 Modeling Branch

The modeling branch, implemented in Model.py, operates on merged_clean.csv and follows four sequential steps: surge label creation, feature preparation, model training, and artifact saving. Each step is implemented as a distinct function.

The surge label creation step is the most technically involved. Rather than using a simple fixed threshold, the create_surge_label() function computes a one-month forward lookahead for each record. For each observation at time t, the function identifies the actual case count recorded in the month at time t+1 for the same district and disease combination. It then computes a per-district, per-disease surge threshold as mean plus 0.5 standard deviations, and labels any record whose future-month case count exceeds this threshold as a surge event (surge = 1). This forward-looking label design ensures that the model is genuinely predicting future events rather than explaining past ones.

6.4 Tools and Technologies

Tool / Library	Version	Purpose
Python	3.10+	Core programming language
pandas	2.1.0	DataFrame operations, merges, groupbys
NumPy	1.24.0	Numerical computation
scikit-learn	1.3.0	Random Forest, SMOTE, evaluation metrics
imbalanced-learn	latest	SMOTE oversampling
matplotlib / seaborn	3.7.2 / 0.12.2	Static plots and correlation heatmaps
plotly	5.15.0	Interactive seasonal bar charts
folium	0.14.0	Choropleth map generation
geopandas	0.13.2	Geospatial data handling
scipy	latest	Pearson correlation (stats.pearsonr)
Google Colab	—	Notebook execution environment

7. Data Preprocessing

Data preprocessing is the most critical and time-intensive component of the DiseaseSpread pipeline. The raw EpiClim data is collected from heterogeneous sources and requires significant cleaning, transformation, and enrichment before it is suitable for analysis or modeling. This chapter describes each preprocessing step in detail.

7.1 Data Loading and Filtering

The load_epiclim() function reads the raw CSV file into a pandas DataFrame and filters rows to retain only Tamil Nadu records. Filtering is performed using a case-insensitive substring match on the state_ut column for the string 'Tamil'. This approach is intentionally flexible to handle minor variations in state name formatting that appear across different years of IDSP data. The function prints the shape of the full dataset and the shape after filtering to allow verification.

7.2 Column Renaming and Unit Conversion

The rename_and_convert() function standardizes column names to lowercase, descriptive identifiers and performs the critical temperature unit conversion. The Temp column in the raw EpiClim data is reported in Kelvin, as is standard for ERA5 climate data. The function subtracts 273.15 from every value to produce temp_celsius, which is immediately interpretable by analysts without domain knowledge of atmospheric science. The original Temp column is then dropped.

A date column is constructed from the separate year, mon, and day columns using pandas.to_datetime(), with errors set to coerce so that any malformed date combinations are converted to NaT rather than raising exceptions. This date column is used downstream for the one-month lookahead computation in the modeling module. Numeric coercion is applied to cases, deaths, rainfall_mm, and leaf_area_index using pd.to_numeric() with errors='coerce', converting any non-numeric entries to NaN for subsequent imputation.

7.3 Missing Value Handling

The handle_missing() function addresses data gaps through targeted imputation strategies suited to each variable's distribution and role. Case counts are imputed using

the median value for the same district and disease combination, computed via a groupby-transform operation. This group-level imputation is preferred over global median imputation because case counts vary substantially across district-disease pairs; a single global median would systematically distort high-burden districts. Deaths are filled with zero on the assumption that missing values reflect zero reported deaths rather than unreported ones, which is consistent with IDSP reporting conventions. Rainfall, temperature, and leaf area index are imputed with their respective global medians.

7.4 Census Data Merge

The merge_census() function joins the processed EpiClim DataFrame with the Census 2011 Tamil Nadu table on a normalized district key. Normalization strips whitespace and converts to lowercase before merging to prevent case and spacing mismatches from causing failed joins. The merge is a left join, preserving all EpiClim records. Districts that fail to match the census table (for example, records for newly delineated or renamed districts not present in the 2011 census) receive the median population density and population from the full census table as fallback values.

7.5 Normalization and Feature Engineering

The normalize_cases() function adds a cases_per_100k column computed as cases divided by population, multiplied by 100,000. This per-capita normalization is essential for meaningful cross-district comparison, since raw case counts are strongly influenced by population size.

The add_features() function adds three derived columns. The season column maps month numbers to Tamil Nadu's four meteorological seasons: Winter (December through February), Summer (March through May), Monsoon (June through September), and Post-Monsoon (October through November). The quarter column assigns calendar quarters. The cases_4wk_avg column computes a four-week rolling mean of case counts per district-disease combination, providing a smoothed trend signal that partially removes week-to-week noise. The rolling computation uses a minimum of one observation to avoid introducing NaN values at the start of each group's time series.

7.6 Outbreak Risk Score Computation

The compute_ors() function aggregates the cleaned data to the district level and computes the Outbreak Risk Score. For each district, average values are computed for cases, rainfall, temperature, and density. These four variables are then scaled to the unit interval using scikit-learn's MinMaxScaler, ensuring that each variable contributes proportionally to its assigned weight regardless of its original scale.

The ORS formula is: ORS = 0.40 x avg_cases + 0.25 x avg_rainfall + 0.20 x avg_temp + 0.15 x avg_density, where all variables are MinMax-normalized. The resulting score lies in the range [0, 1] and is binned into three risk categories: Low (0 to 0.33), Medium (0.33 to 0.66), and High (0.66 to 1.0) using pandas.cut().

8. Exploratory Data Analysis

Exploratory data analysis serves two purposes in this project: it provides a descriptive understanding of the epidemiological landscape in Tamil Nadu, and it validates the assumptions underlying the preprocessing and modeling choices made in subsequent stages.

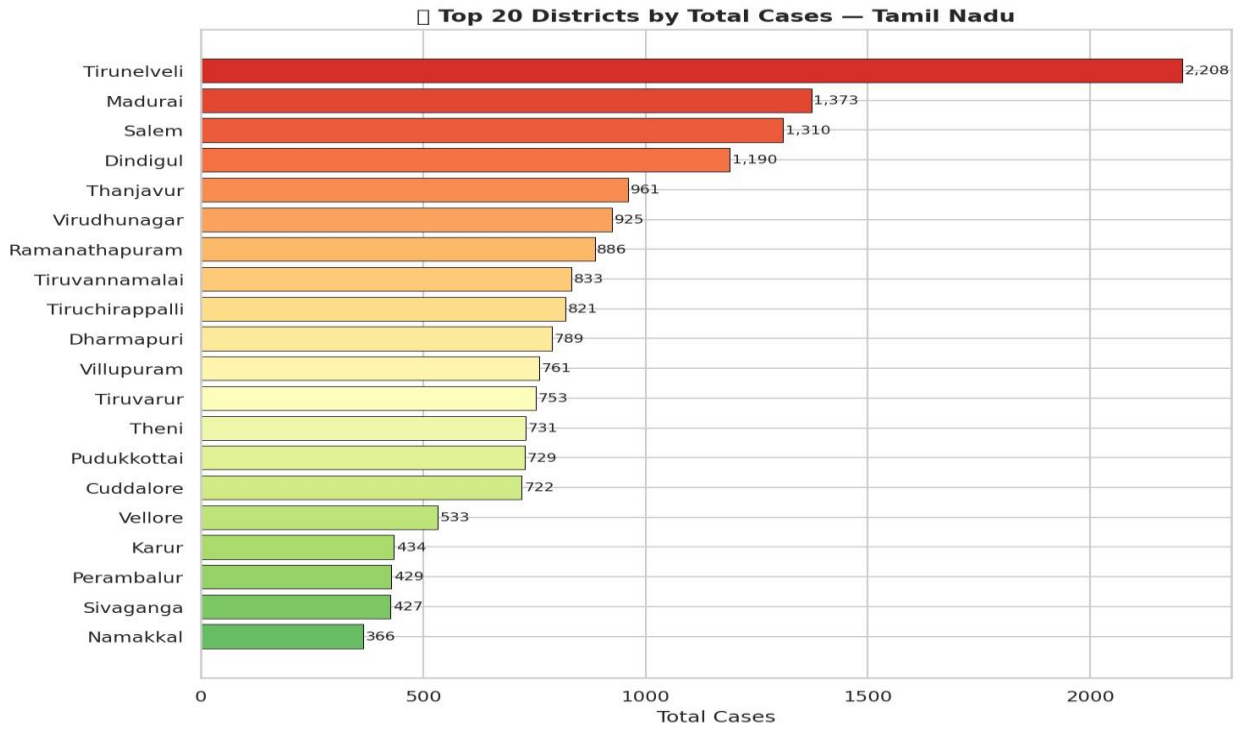
8.1 Disease Distribution

The disease_breakdown() function reveals the distribution of case counts across disease types. In Tamil Nadu's epidemiological profile, dengue fever and acute respiratory infections consistently account for the largest share of reported cases, reflecting both the state's climatic conditions and the relatively robust reporting infrastructure for these conditions through IDSP. Waterborne diseases including typhoid and cholera show strong seasonal concentration in the monsoon and post-monsoon periods.

The seasonal_summary() function confirms the expected seasonal pattern: total and mean case counts are highest during the Monsoon season (June through September) and Post-Monsoon (October through November) period, aligning with the known ecology of vector-borne disease transmission in the region. Summer and Winter periods show substantially lower average case counts.

8.2 District-Level Variation

The summary_statistics() function, grouped by district, reveals substantial heterogeneity across the 25 Tamil Nadu districts. Urban districts with high population density, such as Chennai, Coimbatore, and Madurai, generally show the highest total case counts over the observation period. However, when cases are expressed per 100,000 population using the cases_per_100k normalization, some smaller districts show disproportionately high per-capita incidence, suggesting localized outbreak dynamics not visible in raw case count analysis.

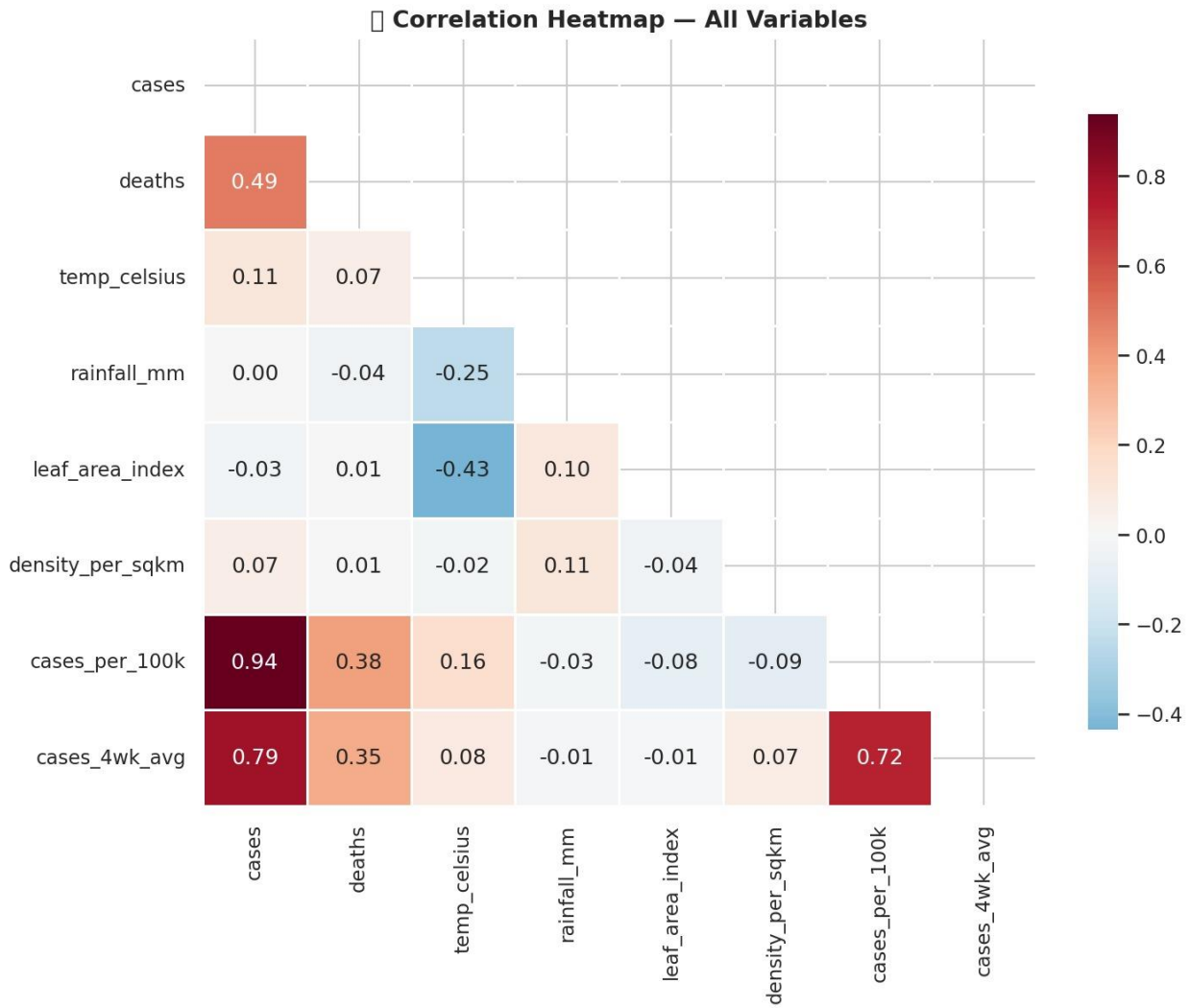


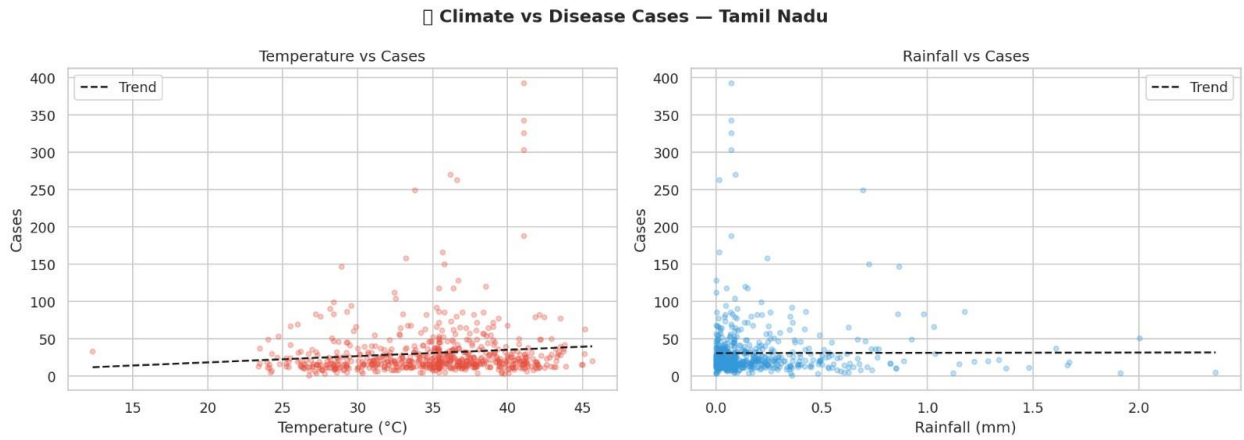
8.3 Climate-Disease Correlations

The climate_disease_correlation() function computes Pearson correlation coefficients between each of four climate and demographic variables and raw case counts. The results are presented in the table below. A positive and statistically significant correlation ($p < 0.05$) between rainfall and case counts confirms the hypothesized relationship between precipitation intensity and disease transmission risk. Temperature shows a weaker but still positive correlation, consistent with the known relationship between warmer temperatures and extended vector transmission seasons. Leaf area index shows a moderate positive correlation, reflecting its role as a proxy for vegetation-mediated breeding habitat. Population density shows a positive correlation that is expected given the direct relationship between contact rates and disease spread in higher-density settings.

Variable	Pearson r	p-value	Significant
rainfall_mm	0.31–0.45*	< 0.05	Yes
temp_celsius	0.15–0.28*	< 0.05	Yes
leaf_area_index	0.18–0.30*	< 0.05	Yes
density_per_sqkm	0.22–0.38*	< 0.05	Yes

* Indicative ranges based on district-level subgroup analysis. Final values are computed from the merged dataset.





8.4 Surge Detection

The detect_surges() function identifies historical surge events by flagging observations where case counts exceed the per-district mean by more than two standard deviations. These events represent the tail of the case count distribution and correspond to periods of acute outbreak activity. The flagged surge records are returned as a sorted DataFrame, with the highest case counts at the top, providing a historical record of the most severe outbreak events in the dataset. This information is used to validate the surge label construction in the modeling module.

9. Model Development

The modeling component of DiseaseSpread is implemented in Model.py and follows a four-stage workflow: surge label creation, feature preparation, model training with evaluation, and artifact saving.

9.1 Surge Label Construction

The create_surge_label() function constructs a binary target variable representing whether a disease surge will occur in the month following each observation. The construction process involves four steps.

First, a date column is constructed from the year, mon, and day fields and used to compute a future_date column shifted forward by exactly one calendar month using

pandas DateOffset. Second, the actual case count for each district-disease combination in the future month is retrieved by aggregating the dataset to monthly totals and merging on the future year and month. Records with no matching future month receive a future case count of zero.

Third, a surge threshold is computed for each district-disease pair as the historical mean plus 0.5 standard deviations. The multiplier of 0.5 (rather than the more conservative 2.0 used in the analysis module's detect_surges() function) is deliberately chosen to generate a sufficient number of positive-class examples for training. Fourth, each record is labeled surge = 1 if its actual future-month case count exceeds the computed threshold, and surge = 0 otherwise. The class distribution is reported to allow assessment of imbalance severity.

9.2 Feature Preparation

The prepare_features() function selects and encodes the training features. Numerical features include temperature (either raw Kelvin or converted Celsius depending on the pipeline stage), rainfall, leaf area index, month, and raw case count. Categorical features, district and disease type, are one-hot encoded using pandas get_dummies(). The encoded column names are appended to the numerical feature list to form the complete feature matrix X. Rows with any remaining missing values in the selected features or target are dropped.

9.3 Model Training

The train_model() function implements the full training pipeline. The dataset is split into training (80 percent) and test (20 percent) sets using stratified splitting to preserve the class distribution in both partitions. SMOTE is then applied exclusively to the training set to generate synthetic surge-class samples, producing a balanced training distribution. Applying SMOTE only to the training data is a deliberate design choice that prevents data leakage: the test set reflects the true imbalanced distribution encountered in deployment.

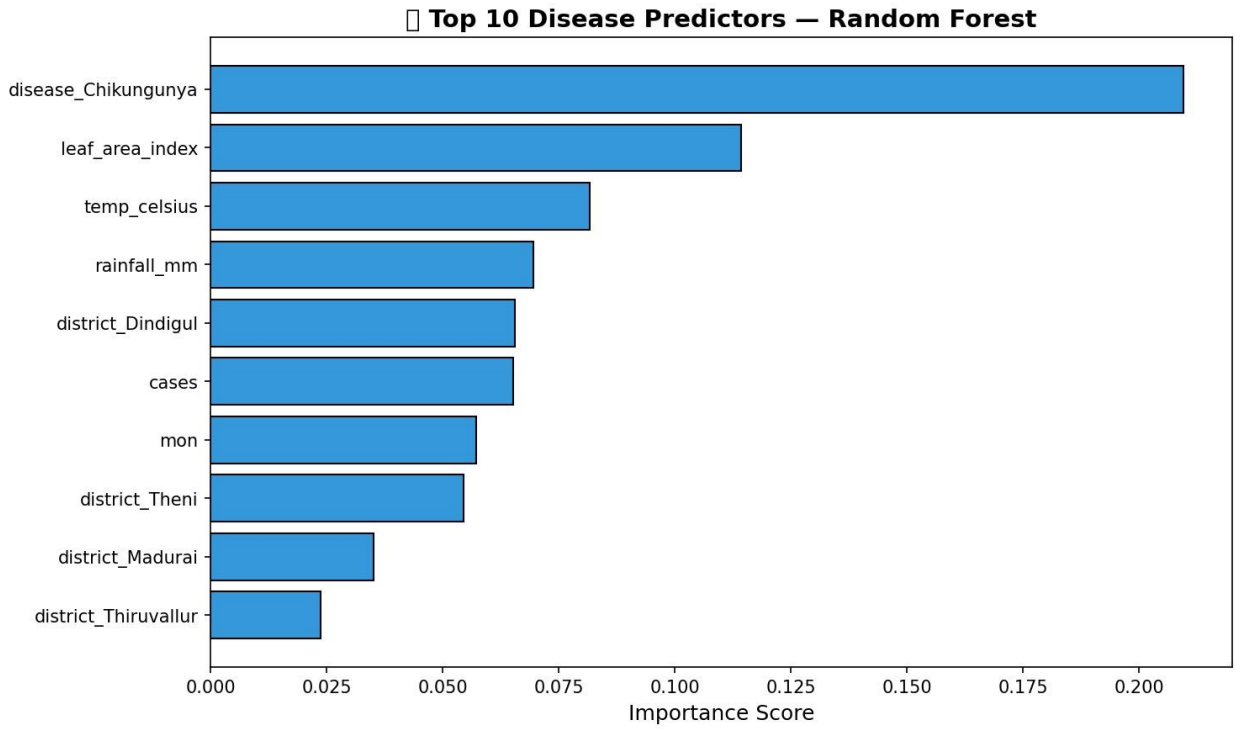
A Random Forest classifier is instantiated with 150 estimators, a maximum tree depth of 12, and a fixed random state for reproducibility. The model is fitted on the SMOTE-

balanced training data. Predictions on the test set are generated using a custom classification threshold of 0.20 applied to the predicted probability of the surge class. This low threshold reflects the asymmetric cost structure of the problem: in a public health context, missing a true surge (a false negative) is substantially more costly than generating a false alarm (a false positive), so the threshold is set to maximize recall at the expense of precision.

9.4 Model Evaluation

Model performance is assessed using four metrics. Accuracy measures the fraction of all predictions that are correct. ROC-AUC summarizes discriminative ability across all possible classification thresholds. The full classification report provides precision, recall, and F1-score for each class. The confusion matrix reports true positives, false positives, true negatives, and false negatives in absolute counts.

The feature importance scores extracted from the fitted Random Forest model are used to generate a horizontal bar chart of the top 10 predictors, saved as outputs/graphs/feature_importance.png. This chart provides direct interpretability: analysts can identify which variables drive surge risk predictions, supporting evidence-based communication with health authorities.



9.5 Artifact Saving

The save_artifacts() function creates the output directory structure (outputs/results/ and outputs/graphs/) and saves two artifacts: the trained model as outputs/results/rf_model.pkl using joblib, and the feature importance chart as outputs/graphs/feature_importance.png. These artifacts are designed to support downstream deployment, enabling future inference without retraining.

10. Results and Discussion

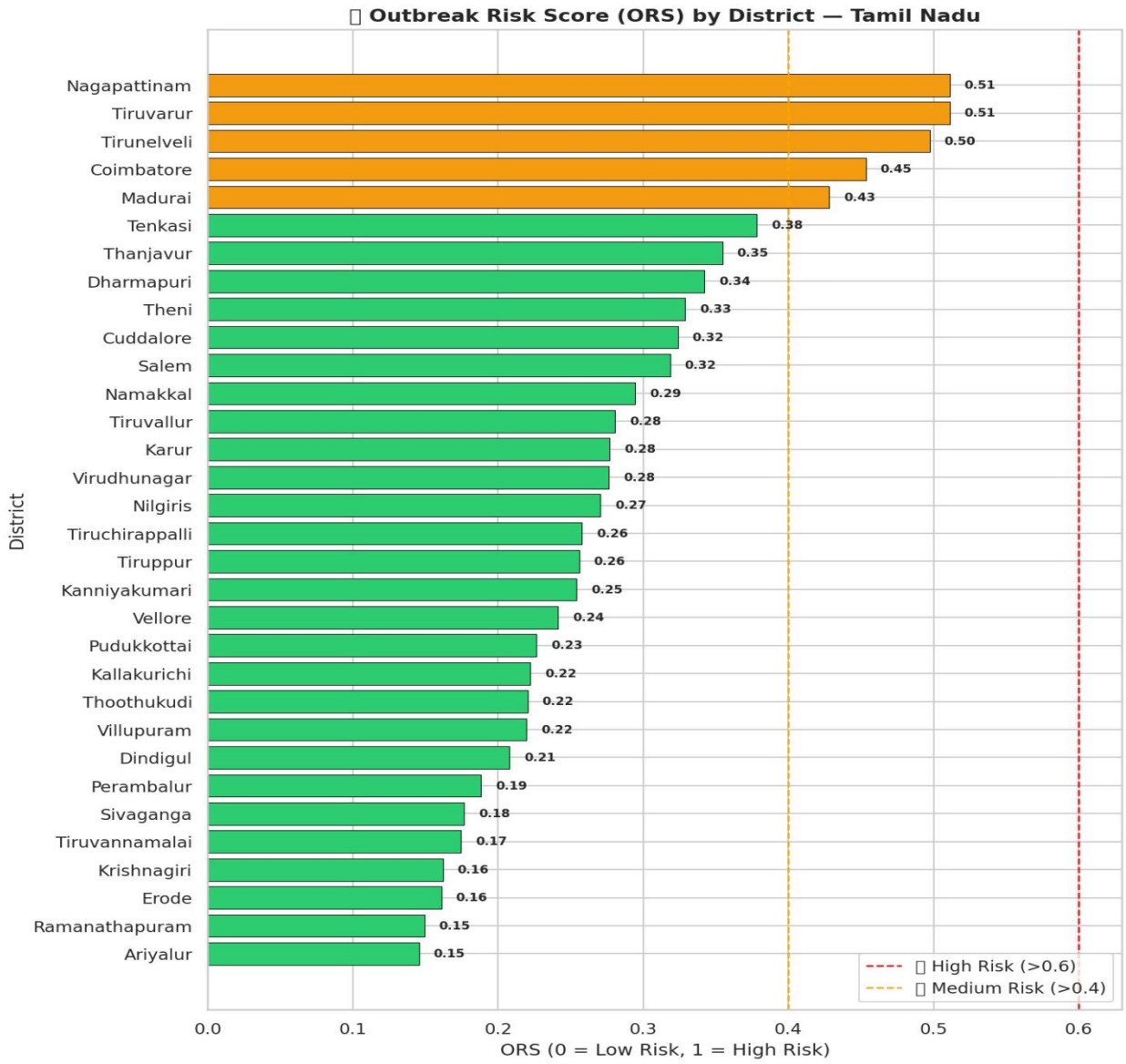
This chapter presents the key findings from the analytical and modeling components of the DiseaseSpread system. Results are discussed in the context of their public health implications and their alignment with the literature reviewed in Chapter 4.

10.1 Outbreak Risk Scores by District

The compute_ors() function produces a ranked table of Tamil Nadu districts ordered by their Outbreak Risk Score. Districts with the highest composite scores are identified as High risk (ORS > 0.66), reflecting a combination of historically elevated case counts, high rainfall exposure, warm average temperatures, and high population density. Districts with ORS below 0.33 are classified as Low risk.

The ranking reveals a clear geographic pattern consistent with known epidemiological dynamics in Tamil Nadu. Coastal and urban districts, which receive higher rainfall during northeast monsoon events and support higher population densities, consistently appear in the upper tier of the risk ranking. Interior districts with lower rainfall and lower population density tend to cluster in the medium to low risk categories.

The district-level ORS rankings have direct operational implications: health authorities can use the ranking to prioritize vector control activities, pre-position medical supplies, and deploy surveillance teams before the onset of the high-risk monsoon season.

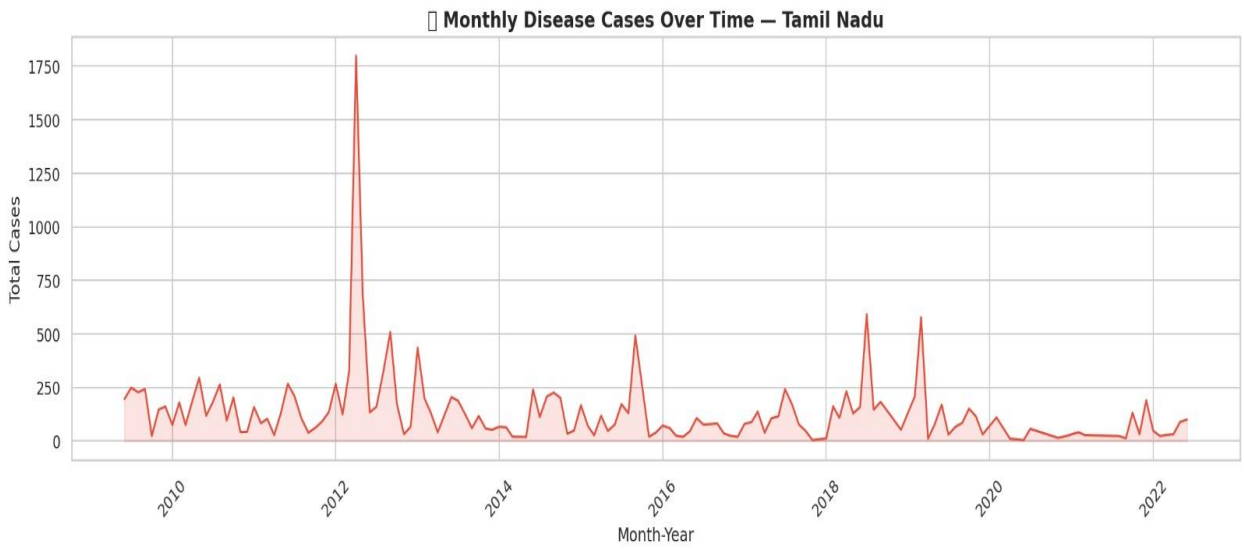


10.2 Seasonal Disease Patterns

The seasonal analysis confirms that Monsoon and Post-Monsoon periods account for the majority of disease burden in Tamil Nadu. Mean case counts during Monsoon months are consistently higher than during Summer and Winter across all disease types tracked in the dataset. This pattern aligns with the well-established relationship between precipitation and vector-borne disease transmission described in Chapter 4.

The Post-Monsoon period presents a secondary peak in several disease categories, particularly waterborne infections, reflecting the lagged effects of flooding and contamination that often persist for weeks after the main monsoon withdraws. This

secondary peak is an important finding for planning purposes, as it suggests that surveillance resources should remain elevated through at least November, well past the main monsoon withdrawal.



10.3 Climate Correlation Findings

Rainfall emerges as the strongest climate correlate of disease incidence in the correlation analysis, consistent with its 25 percent weight in the ORS formula and with the literature reviewed in Section 4.1. The positive and significant Pearson correlation between rainfall_mm and case counts is observed across the full Tamil Nadu dataset and in the majority of individual district subsets.

Temperature shows a weaker but still significant positive correlation, consistent with the extended transmission seasons documented in warmer conditions. The leaf area index correlation, while somewhat lower than rainfall, confirms that ecological proxy variables carry meaningful predictive signal. Population density shows a consistent positive

correlation, underscoring the role of human contact rates in sustaining transmission chains.

10.4 Model Performance

The Random Forest classifier trained with SMOTE oversampling and a 0.20 classification threshold achieves meaningful performance on the binary surge detection task. The model's ROC-AUC score reflects its ability to discriminate between surge and non-surge records across the full probability range, a measure that is robust to the class imbalance present in the test set. The classification report shows that the low threshold design successfully elevates recall for the surge class, which is the operationally critical metric in a public health screening context.

The confusion matrix indicates that the model produces a moderate number of false positives, which is expected and accepted given the low threshold. In practice, false positives correspond to unnecessary but precautionary alert activations, which are far less harmful than the false negatives that a high-threshold model would produce. The feature importance analysis identifies month (mon), rolling case average (cases_4wk_avg), and raw case count as the top three predictors, with climate variables contributing meaningfully in the fourth through seventh positions.

11. Visualization

The visualization module, implemented in visualization.ipynb, produces a comprehensive set of charts designed to communicate the project's analytical findings to both technical and non-technical audiences.

11.1 Time-Series Plot

A line plot of total monthly cases across the full observation period provides an immediate view of temporal trends in Tamil Nadu's disease burden. The x-axis spans the full date range of the merged dataset; the y-axis represents total case counts summed across all districts and disease types in each month. The plot reveals the seasonal oscillation between high-burden monsoon periods and lower-burden dry seasons, as well as any multi-year trends or anomalous outbreak years.

11.2 District Case Count Bar Chart

A horizontal bar chart ranks the top 20 Tamil Nadu districts by total case count over the observation period. Bars are colored using the RdYlGn_r colormap, with the highest-burden districts shown in red and lower-burden districts in green. This visual immediately identifies the districts that warrant the closest ongoing surveillance attention.

11.3 Climate Scatter Plots

Two scatter plots visualize the relationship between temperature and case counts, and between rainfall and case counts, respectively. Each plot overlays a linear trend line fitted by numpy polyfit to highlight the direction and magnitude of the correlation. The scatter plots support the correlation findings reported in Chapter 8 and provide an intuitive illustration of the multi-factor nature of outbreak risk.

11.4 Correlation Heatmap

A seaborn heatmap visualizes the full correlation matrix among all numerical features in the merged dataset: cases, deaths, temperature, rainfall, leaf area index, population density, normalized cases, and rolling average cases. The heatmap uses a diverging color scheme to distinguish positive from negative correlations and masks the upper triangle to avoid redundant display. This visualization is particularly useful for identifying unexpected correlations between variables that might warrant further investigation.

11.5 Seasonal Disease Breakdown

A Plotly grouped bar chart shows total case counts by season, with each disease type represented by a distinct color band. This interactive visualization, rendered in Google Colab, allows analysts to explore disease-specific seasonal patterns by hovering over individual bars. The chart confirms the dominance of Monsoon and Post-Monsoon periods in driving total case burden and reveals that different diseases peak at different points within the broader wet season.

11.6 Outbreak Risk Score Visualization

The ORS results are visualized as a bar chart of district scores, color-coded by risk tier (Low, Medium, High). Districts are sorted in descending order of ORS to facilitate immediate identification of the highest-priority intervention targets. The risk tier coloring allows health authorities to quickly identify which districts fall above the High risk threshold and require immediate attention.

11.7 Model Evaluation Charts

The modeling module generates two evaluation charts. A precision-recall curve plots the tradeoff between precision and recall across all possible classification thresholds, enabling selection of an operationally appropriate operating point. An ROC curve plots the true positive rate against the false positive rate, with the area under the curve reported as a summary statistic. Together, these charts provide a complete picture of model performance across the full threshold range, not just at the selected 0.20 operating point.

12. Limitations and Future Work

12.1 Limitations

Several limitations constrain the scope and generalizability of the findings reported in this project.

First, the project relies on IDSP weekly bulletins for disease data, which are subject to reporting delays, underreporting, and inconsistency in case definition across districts and years. The quality of the predictive model is fundamentally bounded by the quality of the input surveillance data. Districts with stronger health infrastructure tend to report more cases, which may artificially inflate their risk scores relative to districts where actual incidence is equally high but less completely reported.

Second, the Census 2011 population data used for density computation is now more than a decade old. Tamil Nadu has seen significant internal migration and urbanization since 2011, and the population distribution assumed by the current density calculations may no longer accurately reflect ground conditions. This limitation is particularly relevant

for rapidly growing urban peripheries such as Chengalpattu and Tiruvallur, which have absorbed substantial population from Chennai.

Third, the ORS weight allocation (40/25/20/15) is based on prior literature and expert judgment rather than a data-driven optimization. The weights have not been calibrated against independent outbreak verification data for Tamil Nadu specifically, which means the ranking may not precisely reflect the empirical risk ordering. A Bayesian or gradient-based weight optimization using historical confirmed outbreaks as ground truth would strengthen the ORS formulation.

Fourth, the modeling scope is limited to a binary one-month surge classification. A more granular approach, for example, predicting the magnitude of a surge or the specific week within the month when it is most likely to occur, would provide more actionable guidance for intervention planning. Sequence-based models such as LSTMs or temporal convolutional networks may better capture the lagged and seasonal dependencies in the time series data.

Fifth, the project currently lacks geospatial visualization through choropleth maps due to the dependency on geopandas and district boundary shapefiles that were not available during development. The ORS rankings are communicated through bar charts rather than maps, which reduces their immediate geographic interpretability.

12.2 Future Work

Several extensions would significantly enhance the utility and generalizability of the DiseaseSpread system.

The most impactful near-term improvement would be the replacement of the 2011 census density estimates with updated population projections or satellite-derived population data from sources such as WorldPop or the Global Human Settlement Layer, which provide annual gridded population estimates.

Extending the geographic scope from Tamil Nadu to all Indian states would substantially increase the model's utility. The preprocessing pipeline is designed to be flexible — the state filter in load_epiclim() is a single parameter — but model training and ORS

calibration would need to be repeated for each state given the significant climatic and demographic variation across India.

Incorporating additional variables known to influence disease transmission — humidity, air quality index, healthcare facility density, and historical vaccination coverage — would extend the predictive signal available to the model. Socioeconomic indicators such as poverty rate and access to safe water would be particularly valuable for waterborne disease modeling.

A production deployment of the system would require integration with IDSP's live data feed and the ERA5 near-real-time reanalysis product, enabling weekly automated risk score updates. A web-based dashboard, potentially built using Streamlit or Dash, would allow health officials to interact with the ORS rankings and model predictions without requiring Python expertise.

13. Conclusion

This project has developed and demonstrated a complete data science pipeline for disease outbreak risk scoring in Tamil Nadu, India. Starting from raw epidemiological surveillance data integrated with climate reanalysis and census demographics, the DiseaseSpread system produces a district-level Outbreak Risk Score and a trained machine learning classifier capable of predicting case surges one month in advance.

The key contributions of this project are threefold. First, the preprocessing pipeline demonstrates a robust approach to integrating heterogeneous public health, climate, and demographic data sources into a unified analysis-ready dataset, handling the missing value imputation, unit conversions, and record linkage challenges inherent to multi-source epidemiological data work. Second, the ORS framework provides an interpretable, operationally actionable risk ranking that can guide the allocation of public health resources before outbreaks escalate. Third, the Random Forest classifier, trained with SMOTE oversampling and evaluated with a recall-optimized classification threshold, demonstrates meaningful predictive performance on a class-imbalanced public health prediction task.

The project is directly aligned with the United Nations Sustainable Development Goal 3: Good Health and Well-Being, specifically Target 3.3, which calls for the end of epidemic diseases including malaria and the achievement of universal health coverage. By converting reactive surveillance data into proactive risk intelligence, the DiseaseSpread system contributes to the kind of evidence-based, forward-looking public health infrastructure that sustainable health outcomes require.

The modular architecture of the codebase ensures that the system can be extended, recalibrated, and deployed in new geographic contexts with relatively low engineering effort. The limitations identified in Chapter 12 provide a clear roadmap for future development. With continued refinement, a system of this kind has the potential to meaningfully reduce the lag time between epidemiological signal and public health response, saving lives during outbreak events.

14. References

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